AI-Driven Intelligent Railway Operations System

An integrated AI decision-support framework for Sri Lanka

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PP2 COMMENTS & ACTIONS TAKEN

Student ID	Panel Comment	Action Taken
IT22217240	Absent during the panel presentation.	-
IT22051202	Absent during the panel presentation.	-
IT22126610	Improve demand prediction by considering additional influencing factors and validating predictions against current CGR data.	Improved the fraud detection approach and focused on validating ticket fraud detection and risk-aware passenger verification.
IT22151506	Improve demand prediction accuracy through stronger experimental validation and comparison with current CGR analysis.	Focused on improving prediction reliability and comparing model predictions with current CGR analysis/results.

RESEARCH PROBLEM & SOLUTION

Challenges in Sri Lankan Railway Operations

- Unpredictable passenger demand leads to overcrowding and inefficient seat utilization.
- Limited train movement visibility and inaccurate delay predictions affect journey planning.
- Manual ticket verification can overlook suspicious ticketing activities.
- Complex scheduling constraints affect railway operational efficiency.

The CeylonRail Intelligence Solution

- AI-based passenger demand forecasting and adaptive seat allocation.
- Train tracking with machine-learning-based delay prediction.
- Intelligent fraud detection and risk-aware passenger verification.
- Multi-objective train scheduling optimization.

SYSTEM ARCHITECTURE

ECOSYSTEM ARCHITECTURE

Unified AI-Driven Railway Intelligence Platform

01 – Demand Forecasting & Seat Allocation

- Predicts passenger demand and optimizes available seating capacity.

02 – Train Tracking & Delay Prediction

- Processes train movement information and estimates operational delays.

03 – Fraud Detection & Passenger Verification

- Detects suspicious transactions and supports risk-based ticket verification.

04 – Train Scheduling Optimization

- Generates scheduling alternatives based on operational objectives and constraints.

Integrated Architecture:

- Users / Railway Staff → CeylonRail Web Dashboard → Four AI Components → Shared Railway Data Layer → Centralized Database

COMPONENT 01

IT22217240 – NARASINGHE N.K.B.N.N



Multi-Objective Train Scheduling Optimization

- Addresses competing railway scheduling objectives and operational constraints.
- Investigates NSGA-II for multi-objective schedule optimization.
- Considers train delays, congestion, capacity, and resource utilization.
- Evaluates alternative schedules using operational performance indicators.
- Uses relevant demand and delay information to support timetable planning.
- Aims to improve schedule coordination and operational decision-making.

COMPONENT 02

IT22051202 – DIDULANTHA C.S



Real-Time Train Tracking and Delay Prediction

- Processes train locations, timestamps, routes, and timetable information.
- Displays the latest available train movement and operational status.
- Investigates LSTM-based regression for delay prediction.
- Evaluates delay classification using 1,065 test observations.
- Recorded 34.5% classification accuracy, highlighting class imbalance and prediction limitations.
- Provides a foundation for improved tracking and delay prediction with richer operational data.

COMPONENT 03

IT22126610 – MUTHUKUDAARACHCHI V.U



Advanced Intelligent Ticket Fraud Detection and Risk-Aware Passenger Verification

- Analyses 9,800 railway transaction records covering 2020–2024.
- Detects suspicious ticketing behavior, duplicate usage, and unusual transaction patterns.
- Evaluates Isolation Forest, One-Class SVM, and Autoencoder models.
- Autoencoder achieved an F1-score of 0.47, Recall of 0.94, and ROC-AUC of 0.96.
- Categorizes transactions into Low, Medium, and High fraud-risk levels.
- Integrates rule-based ticket validation and passenger verification to support inspector decisions.

COMPONENT 04

IT22151506 – SENAWIRATHNA D.M.N.T



Advanced Spatio-Temporal Demand Forecasting and Adaptive Seat Allocation Using AI

- Analyses 55,302 historical railway records from 2020–2024.
- Evaluates Random Forest, XGBoost, and LSTM forecasting approaches.
- Extracts temporal, seasonal, route, and passenger-class demand patterns.
- Random Forest achieved an MAE of 24.10, RMSE of 37.67, and MAPE of 13.12%.
- Uses integer optimization for adaptive seat allocation.
- Supports demand-aware railway capacity planning.

SYSTEM EVALUATION

Model Performance and Experimental Findings

Component	Evaluation Result
Demand Forecasting	Random Forest – MAPE: 13.12%
Train Delay Classification	Accuracy: 34.5%
Fraud Detection – Isolation Forest	F1-score: 0.36
Fraud Detection – One-Class SVM	F1-score: 0.46
Fraud Detection – Autoencoder	F1-score: 0.47
Train Scheduling Optimization	Final individual results pending

Key Findings

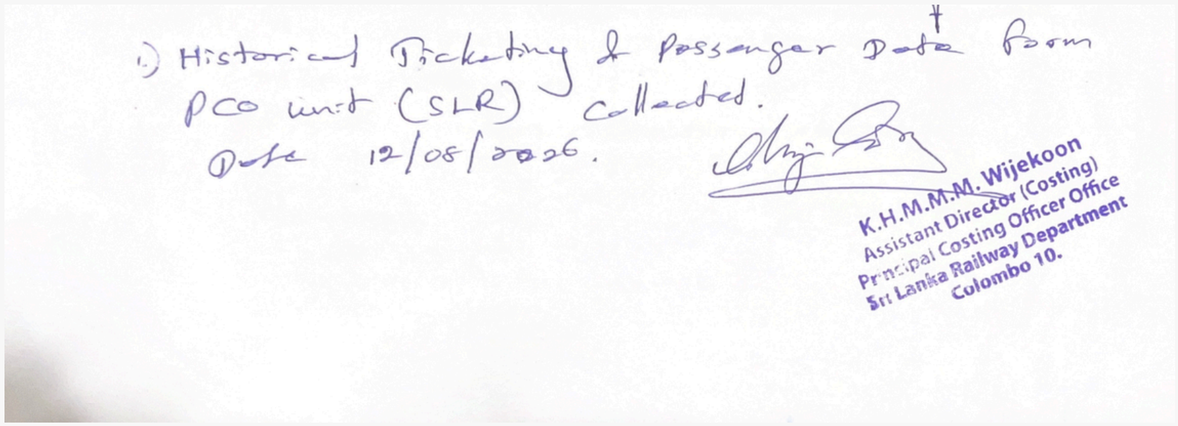
- Demand forecasting supports data-driven capacity allocation.
- Fraud detection models demonstrate strong recall but require improved precision.
- Delay classification needs further improvement to distinguish delay categories.
- Integrated railway intelligence provides a foundation for coordinated decision support.

LIMITATIONS & FUTURE WORK

Limitations and Future Improvements

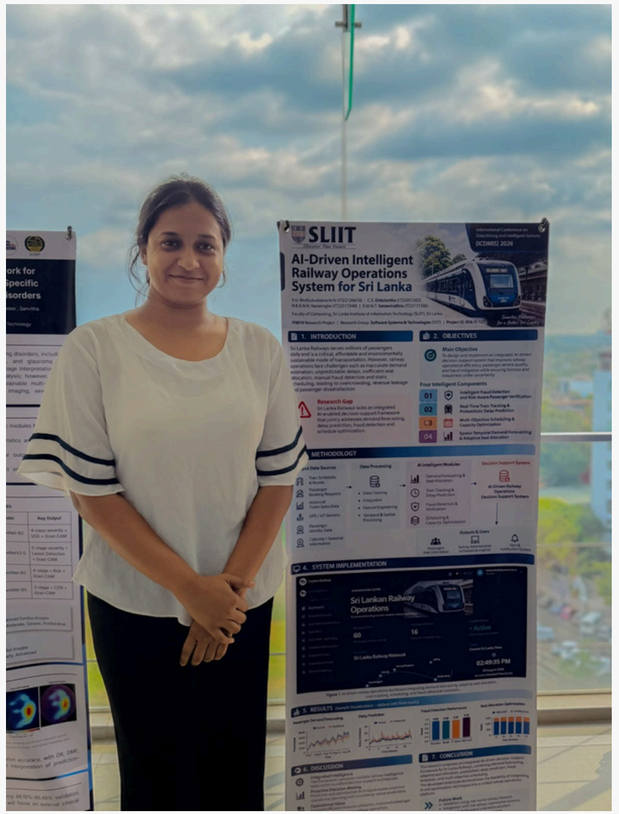
- Limited availability of real-time Sri Lankan railway operational data.
- Machine-learning performance depends on dataset quality and representativeness.
- Fraud detection may produce false positives requiring manual verification.
- Delay prediction requires improved class-sensitive model performance.
- Scheduling recommendations require further operational feasibility validation.
- Future work includes larger datasets, improved models, live data integration, and railway stakeholder testing.

User Feedback & Publications



i) Historical Ticketing & Passenger Data Form PCO unit (SLR) collected.
Date 12/05/2026.

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CONCLUSION

CEYLONRAIL INTELLIGENCE

Smarter Railways. Better Decisions. Intelligent Operations.

CeylonRail Intelligence demonstrates an integrated AI-driven approach to railway demand forecasting, train tracking, ticket fraud detection, and scheduling optimization.

The research provides a foundation for more informed, efficient, and coordinated railway decision-making in Sri Lanka.

Thank You!

Research Group – R26-IT-127

Sri Lanka Institute of Information Technology

Faculty of Computing